NOT MEASUREMENT SENSITIVE

MIL-STD-2045-14503
19 April 1995

MILITARY STANDARD

INFORMATION TECHNOLOGY
DOD STANDARDIZED TRANSPORT PROFILE
INTERNET TRANSPORT SERVICE SUPPORTING OSI APPLICATIONS

AMSC N/A

AREA DCPS

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:40Z
Check the source to verify that this is the current version before use.

Foreword

This military standard is approved for use by all Departments and Agencies of the Department of Defense (DOD).

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this MIL-STD should be addressed to the:

Joint Interoperability and Engineering Organization (JIEO)
ATTN: JEBBF
Fort Monmouth, New Jersey 07703-5613

by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this MIL-STD or by memorandum.

This MIL-STD-2045-14503 series DOD Standardized Profile (DSP) is a functional standard produced by the Data Communications Protocol Standards (DCPS) Technical Management Panel (DTMP). DTMP functional standards are functional groupings of base standards. Referenced base standards may be commercial, DOD or de facto standards, although International Standards (produced by ISO, CCITT, and other bodies) are preferred when possible.

This Defense Standardized Profile (DSP) is a functional DOD Data Communications Protocol Standard (DCPS) produced by the DCPS Technical Management Panel (DTMP). The MIL-STD-2045 document series was established within the DCPS Standardization Area to allow for the enhancement of commercial standards or the development of standards that are unique to DOD.

The MIL-STD-2045-10000 series, MIL-STD-2045-10000 to MIL-STD-2045-19999 inclusive, will be used to describe how DOD will implement commercial, international, national, federal, or military standards within the functional profile concept to provide required network services. The Government Open Systems Interconnection Profiles (GOSIP) will serve as the base for developing the 10000 series with DOD enhancements, unique military standards, and interim standards being used only when necessary.

The MIL-STD-2045-20000 series, MIL-STD-2045-20000 to MIL-STD-2045-29999 inclusive, will be used to describe DOD enhancements and extensions to existing commercial, international, national, or federal standards.

The MIL-STD-2045-30000 series, MIL-STD-2045-30000 to MIL-STD-2045-39999 inclusive, will be used to describe protocols and services unique to DOD that will not be supported by commercial, international, national, or federal standards.

The MIL-STD-2045-40000 series, MIL-STD-2045-40000 to MIL-STD-2045-49999 inclusive, will be used to document interim standards. Interim standards document protocols and services needed by DOD until these protocols and services are described in either a GOSIP or a MIL-STD-2045-20000 or -30000 series standard.

The MIL-STD-2045-50000 series, MIL-STD-2045-50000 to MIL-STD-2045-59999 inclusive, will be used to describe how DOD will implement commercial, international, national, federal, or military standards within the functional profile concept to provide required network services. The Government Open Systems Interconnection Profiles (GOSIP) will serve as the base for developing the 50000 series with DOD enhancements, unique military standards, and interim standards being used only when necessary. The difference between MIL-STD-2045-10000 series and the MIL-STD-2045-50000 series is that the 50000 series are interim profiles.

Specific details and instructions for establishing a MIL-STD-2045 document, as well as profile development guidelines, are documented in MIL-HDBK-829. DTMP Working Groups shall be responsible for DSP development and informal Service or Agency coordination; the DTMP Plenary shall be responsible for final review and approval.

Source: https://assist.dla.mil -- Downloaded: 2026Ю-24T01:40Z
Check the source to verify that this is the current version before use.

This document is part of a set of interim DOD data communications protocol profiles based on the Internet protocol suite and is intended to support the interoperability of DOD communication networks, including connectivity with the Defense Data Network (DDN).

This part of MIL-STD 2045-14503 contains one normative annex and one informative annex:

Annex A (normative) DSPICS REQUIREMENTS LIST (DPRL).
Annex B (informative) CONCLUDING MATERIAL.

For DOD acquisition purposes, where such differences exist, this DSP shall be the controlling document.

The Preparing Activity for this standard is the Data Communication Protocol Standards Technical Management Panel (DTMP). The custodians for the document are identified in the Defense Standardization Program, "Standardization Directory (SD-1)" and are classified in the Federal Supply Classification (FSC) system under Data Communication Protocol Standards (DCPS). Additional information can be obtained from:

Joint Interoperability and Engineering Organization
ATTN: DTMP Chairman
Ft. Monmouth, New Jersey 07703-5613

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:40Z
Check the source to verify that this is the current version before use.

Contents

Introduction		vi
1	Scope	1
1.1	General	1
1.2	Position Within the Taxonomy	1
1.3	Scenario	1
2	References	2
2.1	Government Documents:	2
2.1.1	Specifications, standards, and handbooks	2
2.1.2	Other Government documents, drawings, and publications	2
2.2	Non-Government publications	2
2.2.1	Profiles.	2
2.2.2	Base Standards	2
2.2.3	Other Non-Government documents, drawings, and publications.	3
2.3	Order of precedence.	4
3	Definitions	4
4	Abbreviations and Acronyms	4
5	Requirements	5
5.1	General Requirements	5
5.2	Conformance Requirements	5
5.2.1	Transport Layer Requirements	5
5.2.2	Transmission Control Protocol (TCP)	5
5.2.3	ISO Transport Protocol Class 0 (TP0)	6
5.2.3.1	TP0 General Requirements	6
5.2.3.2	TP0 Detailed Requirements	6
5.2.3.2.1	TPO Elements of procedures	6
5.2.3.2.1.1	Assignement to network connection	6
5.2.3.2.1.2	TPDU transfer	6
5.2.3.2.1.3	Segmenting and reassembling	6
5.2.3.2.1.4	Concatenation and separation	7
5.2.3.2.1.5	Connection establishment	7
5.2.3.2.1.6	Connection refusal	7
5.2.3.2.1.7	Normal release	7
5.2.3.2.1.8	Error Release	7
5.2.3.2.1.9	Association of TPDUs with transport connection	7
5.2.3.2.1.10	TPDU numbering	7
5.2.3.2.1.11	Expedited data transfer	7
5.2.3.2.1.12	Explicit flow control	8
5.2.3.2.1.13	Checksum	8
5.2.3.2.1.14	Protocol Errors	8
5.2.3.2.2	Specification for class 0: Simple class	8
5.2.3.2.2.1	Functions of class 0	8
5.2.3.2.2.2	Procedures for class 0	8
5.2.3.2.2.3	Data transfer	8
5.2.3.2.2.4	Release	8
5.2.3.2.3	Structure and encoding of TPDUs	9
5.2.3.2.3.1	CR: connection request	9
5.2.3.2.3.2	CC: connection confirm	9

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the current version before use.

5.2.3.3.3 DR: disconnect request . . . 9
5.2.3.3.4 DT: data . . . 9
5.2.3.2.3.5 ED: expedited data . . . 9
5.2.3.2.3.6 ER: TPDU error . . . 9
5.2.3.2.3.7 Network Connection Management (NCMS) . . . 9
5.2.3.2.3.8 Class negotiation . . . 9
5.2.3.2.3.9 Size negotiation . . . 10
5.2.3.2.3.10 Packet Format . . . 10
5.2.3.3 Network Services . . . 10
5.3 Interface between TP0 and TCP . . . 10
5.3.1 TCP Service Primitives . . . 10

Annexes

A DSPICS REQUIREMENTS LIST (DPRL) . . . A-1
A.1 Introduction . . . A-1
A.1.1 Notation . . . A-1
A.1.2 Footnotes . . . A-3
A.1.3 Instructions for Completing the DPRL . . . A-3
A.2 Standards Referenced . . . A-3
A.3 DSPICS Requirements List . . . A-4
A.3.1 General Information . . . A-4

B CONCLUDING MATERIAL . . . B-1
B.1 Deviations from the Base Standards/Referenced profiles . . . B-1
B.2 Subject Term (Key word) Listing . . . B-1
B.3 Preparing Activity . . . B-1
B.4 Reviewing Activities . . . B-1
B.5 Custodians . . . B-2

Tables

A.3.1.1 Implementation Identification . . . A-4
A.3.1.2 Identification of IUT . . . A-4
A.3.1.3 Global Statement of Conformance . . . A-4
A.4 Basic Requirements . . . A-5
A.4.1 PICS Proforma for TP0 (modified IAW RFC 1006) . . . A-5
A.4.1.1 Identification . . . A-5
A.4.1.2 Protocol Summary . . . A-5
A.4.1.3 Classes Implemented . . . A-6
A.4.1.3.1 Initiator /responder capability for protocol classes 0-4 . . . A-6
A.4.1.4 Supported Functions for class 0 (C0::) . . . A-6
A.4.1.4.1 Supported TPDUs . . . A-7
A.4.1.4.2 Support of ER on transmission . . . A-7
A.4.1.4.3 Supported parameters for class 0 TPDUs . . . A-8

Figures

Figure 1 OSI/TCP Scenario . . . 1

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the currentversion before use.

Introduction

This DOD Standardized Profile (DSP) is defined within the context of functional standardization, in accordance with the principles specified by ISO/IEC TR 10000, "Framework and Taxonomy of International Standardized Profiles" and MIL-HDBK-829. The context of functional standardization is one part of the overall field of Information Technology (IT) standardization activities - covering base standards, profiles, and registration mechanisms. A profile defines a combination of base standards that collectively perform a specific well-defined IT function. Profiles standardize the use of options and other variations in the base standards to promote system interoperability and to provide a basis for the development of uniform, internationally recognized system tests.

One of the most important roles for a DSP is to serve as the basis for the development of recognized tests. DSPs also guide implementors in developing systems that fit the needs of the US Department of Defense (DoD). DSPs are produced not simply to 'legitimize' a particular choice of base standards and options, but to promote real system interoperability. The development and widespread acceptance of tests based on this and other DSPs is crucial to the successful realization of this goal.

The base standards of this DSP include Request For Comments (RFCs) designated as Official Internet Architecture Board (IAB) standards, other RFCs, and Open Systems Interconnection (OSI) Layer Standards from the OSI Reference Model.

The specifications in this part of DOD Standardized Profile (DSP) 2045-14503 cover the OSI Applications on top of TCP (OSI/TCP). The current technical content of this document has been derived wherever possible from Internet Architecture Board Request For Comments (RFC) 1006. This document must be combined with: ISO/IEC 8073, Open Systems Interconnection Protocol For Providing the Connection Mode Transport Service; IAB STD 7, Transmission Control Protocol (TCP) and IAB STD 5, Internet Protocol.

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the currentversion before use.

1 Scope

1.1 General

This part of DOD Standardized Profile (DSP) 2045-14503 applies to the OSI Applications on top of TCP (OSI/TCP).

1.2 Position Within the Taxonomy

This profile is classified as MIL-STD 2045-14503 in accordance with MIL-HDBK 829 Volume 1.

1.3 Scenario

This DSP specifies the provisions of the OSI Applications on top of TCP using TP0 (OSI/TCP).

<table>
<tr><td>ISO TRANSPORT LAYER</td><td>ISO/IEC 8073
TP-0
IAB-STD 35 OSI over TCP
(RFC 1006)</td></tr>
<tr><td>INTERNET TRANSPORT LAYER</td><td>IAB STD 7,TCP
(RFC 793)
(MIL-STD-2045-14502)</td></tr>
<tr><td>NETWORK LAYER</td><td>IAB STD 5, IP
(RFCs 791,792 919, 922, 1112)
(MIL-STD-2045-14502)</td></tr>
<tr><td>DATA LINK LAYER</td><td rowspan="2">SUBNETWORK DEPENDENT
(MIL-STD-2045-14502)</td></tr>
<tr><td>PHYSICAL LAYER</td></tr>
</table>

Figure 1 OSI/TCP Scenario

2 References

2.1 Government

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the currentversion before use.

Documents:

2.1.1 Specifications, standards, and handbooks

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

MIL-HDBK 829: July 1994 *Guidelines for Developing Data Communications Protocol Standards*

MIL-STD-2045Version 1.0, Part 1, 1994-02-10 *(Information Technology - Defense Standardized Profile (DSP) - Internet Transport Profiles - Part 1: Transport Services)*

DoD activities may obtain copies of DoD directives through their own publication channels or from the DoD Single Stock Point, Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Other federal agencies and the public may purchase copies from the U.S. Department of Commerce, National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161.

Copies of Federal Information Processing Standards (FIPS) are available to Department of Defense activities from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099. Others must request copies of FIPS from the National Technical Information Services, 5285 Port Royal, Springfield, VA 22161-2171.

2.1.2 Other Government documents, drawings, and publications

None

2.2 Non-Government publications

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solication.

2.2.1 Profiles.

None

2.2.2 Base Standards.

IAB STD-35, ISO Transport Service on top of the TCP

ISO/IEC 8073 *(1992-12-15, Information technology Telecommunications and information exchange between systems- Open Systems Interconnection- Protocol for providing the connection-mode Transport Service)*

ISO/IEC 8072 *(1986, Information processing systems- Open systems Interconnection- Transport Service definition)*

ISO 7498-2: 1990, *Information processing systems - Open Systems Interconnection - Basic Reference Model - Part 2: Security Architecture.*

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the currentversion before use.

RFCs are in the public domain and are available on the Internet

ISO Documents may be obtained from the American National Standards Institute, 11 West 42nd Street, NY, NY 10036 or to ISO, Van Demonstrate 94, 1013 CN Amsterdam, Netherlands.)

2.2.3 Other Non-Government documents, drawings, and publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solication.

ISO/IEC 7498	*(1984, Information processing systems- Open systems Interconnection-Basic Reference Model)*
ISO/IEC 7498	*(1984/Add.1:1987, information processing systems- Open systems Interconnection-Basic Reference Mode- Addendum 1: Connectionless-mode transmission)*
ISO/IEC 7498	*(1989, Information processing systems- Open systems Interconnection-Basic Reference Model-Part3: Naming and addressing)*
ISO/IEC 8348	*(1992, Information processing systems- Open systems Interconnection- Data communications- Network service definition)*
ISO/IEC 9646-1	*(1991, Information technology- Open systems Interconnection- Conformance testing methodology and framework- Part1: General concepts)*
ISO/IEC 9646-2	*(1991, Information technology- Open systems Interconnection- Conformance testing methodology and framework- Part: Abstract test suite specification)*
ISO/IEC 11570	*(1992, Information technology- Telecommunications and information exchange between systems- Open Systems Interconnection-Transport protocol identification mechanism)*
CCITT X.224	*(1988, Transport Protocol Specification for Open systems Interconnection for CCITT Applications Version 1988)*
CCITT X.214	*(1984, Transport Service Definitions for Open systems Interconnection (OS) for CCITT Applications)*
IAB STD-5,	*(Internet Protocol)*
IAB STD-7,	*(Transmission Control Protocol)*

2.3 Order of precedence.

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 Definitions

For the purposes of this part of MIL-STD 2045-14503, the following definitions apply.

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the currentversion before use.

Terms used in this part of MIL-STD 2045-14503 are defined in the referenced base standards. Additionally, the following terms are defined.

Internet Architecture Board (IAB) Standards (STD): The IAB has established this as an official standard protocol for the Internet. These protocols are assigned STD numbers.

Request For Comments (RFCs): RFC are the working notes of the "Network Working Group"; that is, the Internet research and development community.

Note: All standards are published as RFCs, but not all RFCs specify standards.

4 Abbreviations and Acronyms

ACK	ACKnowledgement
ALP	Application Layer Protocol
DoD	Department Of Defense
DSP	Defense Standardized Profile
GOSIP	Government Open Systems Interconnection Profile
IAB	Internet Architecture Board
IAB	Internet Architecture Board
ICMP	Internet Control Message Protocol
IHL	Internet Header Length
IP	Internet Protocol
IP	Internet Protocol
ISN	Initial Sequence Number
MSS	Maximum Segment Size
RFC	Request For Comments
STD	Standard
SYN	Synchronization
TC	Transport Connection
TCEP	Transport Connection End Point
TCP	Transmission Control Protocol
TCP	Transmission Control Protocol
TOS	Type Of Service
TTL	Time To Live
DPRL	DSPICS Requirements List
DSPICS	Defense Standardized Profile Implementation Conformance Statements
IEC	International Electrotechnical Commission,
ISO	International Organization for Standardization
IUT	Implementation under test
MIL	Military
PICS	Protocol Implementation and Conformance Statement
QOS	Quality of Service
TP0	Transport Protocol Class 0
TPDU	Transport Protocol Data Unit
TS	Transport Service
TSDU	Transport Service Data Unit

5 Requirements

5.1 General Requirements

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the currentversion before use.

Implementations claiming conformance to this DSP shall support the following elements as stated.

(a) All requirements in the remainder of section 5.
(b) All mandatory requirements of the standards referenced by this profile.
(c) All of the constraints specified in Annex A (normative).

5.2 Conformance Requirements

Implementations claiming conformance to this DSP 2045-14503 shall support the following as stated

5.2.1 Transport Layer Requirements

This profile is used to obtain OSI end to end services over TCP/IP based networks. The profile specifies TP0 on top of TCP. TP0 functions as the transport service convergence protocol and provides a packet orientation on top of the TCP octet stream while using the end-to-end services of TCP. The TPDUs exchanged by the TP0 layer are delimited when passing through the TCP service interface.

5.2.2 Transmission Control Protocol (TCP)

A conforming implementation of this profile shall support Transport Layer requirements as specified in clause 5.2 of Mil-Std 2045-14502, para 5.2.

5.2.3 ISO Transport Protocol Class 0 (TP0)

A conforming implementation of this profile shall support the following capabilities in accordance with ISO/IEC 8073 with the exception of the following modifications that are specified in RFC 1006. Modifications include some QOS parameters. Initial data may be exchanged during connection establishment. Expedited data service is supported, a larger TSDU size is supported, and the network service is supported by TCP. Only TPDUs formatted for class 0 will be exchanged.

5.2.3.1 TP0 General Requirements

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073 Section one, PARA 5.4.1

5.2.3.2 TP0 Detailed Requirements

This paragraph identifies additional requirements, or clarifications of requirements to support military use of TP0 on top of TCP.

5.2.3.2.1 TPO Elements of procedures

5.2.3.2.1.1 Assignment to network connection

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073, Section two, para 6.1.1.

5.2.3.2.1.2 TPDU transfer

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the current version before use.

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073 Section two, para. 6.2.

5.2.3.2.1.3 Segmenting and reassembling

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073 Section two, para. 6.3.

5.2.3.2.1.4 Concatenation and separation

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073 Section two, para. 6.4.

5.2.3.2.1.5 Connection establishment

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073, Section two, para. 6.5.

5.2.3.2.1.6 Connection refusal

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073, Section two, para. 6.6.

5.2.3.2.1.7 Normal release

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073 ,Section two, para. 6.7.

5.2.3.2.1.8 Error Release

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073 ,Section two, para. 6.8.

5.2.3.2.1.9 Association of TPDUs with transport connection

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073 ,Section two, para. 6. 6.9.

5.2.3.2.1.10 TPDU numbering

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073 ,Section two, para. 6.10.

5.2.3.2.1.11 Expedited data transfer

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the currentversion before use.

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073 ,Section two, para. 6.11.

5.2.3.2.1.12 Explicit flow control

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073 ,Section two, para. 6.16.

5.2.3.2.1.13 Checksum

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073 ,Section two, para. 6.17.

5.2.3.2.1.14 Protocol Errors

There are no additional requirements to the TP0 General Requirements as specified in ISO/IEC 8073 ,Section two, para. 6.22.

5.2.3.2.2 Specification for class 0: Simple class

There are no additional requirements to TP0 as specified in ISO/IEC 8073 ,Section two, para. 8.

5.2.3.2.2.1 Functions of class 0

There are no additional requirements to TP0 as specified in ISO/IEC 8073 ,Section two, para. 8.1

5.2.3.2.2.2 Procedures for class 0

There are no additional requirements to TP0 as specified in ISO/IEC 8073 ,Section two, para 8.2

5.2.3.2.2.3 Data transfer

There are no additional requirements to TP0 as specified in ISO/IEC 8073 ,Section two, para 8.2.3, RFC 1006 para 5, pg.13.

5.2.3.2.2.4 Release

There are no additional requirements to TP0 as specified in ISO/IEC 8073 Section 8.2.4 and RFC 1006 para 5, pg. 13.

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the currentversion before use.

5.2.3.2.3 Structure and encoding of TPDUs

There are no additional requirements to TP0 as specified in ISO/IEC 8073 ,Section two, para. 13.

5.2.3.2.3.1 CR: connection request

There are no additional requirements to TP0 as specified in ISO/IEC 8073 ,Section two, para. 13.3.

5.2.3.2.3.2 CC: connection confirm

There are no additional requirements to TP0 as specified in ISO/IEC 8073 ,Section two, para. 13.4.

5.2.3.2.3.3 DR: disconnect request

There are no additional requirements to TP0 as specified in ISO/IEC 8073 ,Section two, para. 13.5.

5.2.3.2.3.4 DT: data

There are no additional requirements to TP0 as specified in ISO/IEC 8073 ,Section two, para. 13.6.

5.2.3.2.3.5 ED: expedited data

There are no additional requirements to TP0 as specified in ISO/IEC 8073 ,Section two, para. 13.8 and RFC 1106 Section 6 pg. 15.

5.2.3.2.3.6 ER: TPDU error

There are no additional requirements to TP0 as specified in ISO/IEC 8073 ,Section two, para 13.12.

5.2.3.2.3.7 Network Connection Management (NCMS)

NCMS is not implemented , network management is accomplished by TCP as per RFC 1006 para. 5. Pg. 13.

5.2.3.2.3.8 Class negotiation

Class negotiation is not implemented, TP0 is the only class supported.

5.2.3.2.3.9 Size negotiation

Size is 65531 octets as specified in RFC 1006 para. 5, pg.14.

5.2.3.2.3.10 Packet Format

TPDUs shall be delimited using the packet format described in RFC 1006 para. 6, pg.14.

5.2.3.3 Network Service

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the currentversion before use.

Network Service shall be provided by TCP as specified in RFC 1006 ,para. 5.

5.3 Interface between TP0 and TCP

5.3.1 TCP Service Primitives

TCP service primitives shall be mapped to the events/actions described in RFC 1006 ,para. 4 and 5

5.4 Details Conformance

A system claiming to implement the procedures specified in this profile shall comply with the requirements specified in ISO/IEC 8073 Section two, para 14 for TP0 except as modified above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:40Z
Check the source to verify that this is the current version before use.

ANNEX A (normative)
DSPICS REQUIREMENTS LIST (DPRL)

A.1 Introduction

This document provides the DoD Standardized Profile Implementation Conformance Statements (DSPICS) Requirements List (DPRL) for implementations of the DoD Standardized Profile (DSP) 2045-14503 The DSPICS for an implementation is generated by completing the DPRL in accordance with the following instructions.

An implementation shall satisfy the mandatory conformance requirements of the base standards referenced in this profile.

An implementation's completed DPRL is called the DSPICS. The DSPICS states which capabilities and options of the protocol have been implemented. The following can use the DSPICS:

(a) the protocol implementor, as a checklist to reduce the risk of failure to conform to the standard through oversight.

(b) the supplier and acquirer or potential acquirer of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard DSPICS proforma.

(c) the user or potential user of the implementation, as a basis for initially checking the possibility of inter-working with another implementation (note that, while inter-working can never be guaranteed, failure to internetwork can often be predicted from incompatible DSPICSs).

(d) a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

A.1.1 Notation

The following notations and symbols from MIL-HDBK 829, which references ISO/IEC TR 10000-1 and -2, are used in the DPRL to indicate the status of features:

Status Symbols

m	- mandatory.
m.<n>	- support of every item of the group labeled by the same numeral <n> required, but only one is active at a time.
o	- optional.
o.<n>	- optional, but support of at least one of the group of options labeled by the same numeral <n> is required.
c	- conditional.
-	- non-applicable (i.e. logically impossible in the scope of the profile).
x	- excluded or prohibited.
i	- out of scope of profile (left as an implementation choice).

In addition, the symbol "•" is used to indicate an option whose status is not constrained by the profile (status in the base standard). The o.<n> notation is used to show a set of selectable options (i.e., one or more of the set must be implemented) with the same identifier <n>.

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the currentversion before use.

Two character combinations may be used for dynamic conformance requirements. In this case, the first character refers to the static (implementation) status, and the second refers to the dynamic (use); thus "mo" means "mandatory to be implemented, optional to be used."

Notations for Conditional Status

The following predicate notations are used:

<predicate>:: This notation introduces a group of items, all of which are conditional on <predicate>.

<predicate>: This notation introduces a single item which is conditional on <predicate>.

In each case, the predicate may identify a profile feature, or a boolean combination of predicates. ("^" is the symbol for logical negation.)

<index>: This predicate symbol means that the status following it applies only when the DPICS states that the features identified by the index are supported. In the simplest case, <index> is the identifying tag of a single DPICS items. The symbol <index> also may be a Boolean expression composed of several indices.

<index>:: When this group predicate is true, the associated clause should be completed.

Notations used in the Protocol Feature Column

<r> Symbol used to denote the receiving system.
<t> Symbol used to denote the transmitting system.

Support Column Symbols

The support of every item as claimed by the implementor is stated by circling the appropriate answer (Yes, No, or N/A) in the support column:

Yes Supported by the implementation.
No Not supported by the implementation.
N/A Not applicable.

Base standard requirements are shown using the equivalent notations in upper case (e.g., M, O, X).

A.1.2 Footnotes

Footnotes to the proforma are indicated by superscript numerals. The footnote appears on the page of the first occurrence of the numeral. Subsequent occurrences of a numeral refer to the footnote of the first occurrence.

A.1.3 Instructions for Completing the DPRL

A DSP implementor shows the extent of compliance to a DSP by completing the DPRL; that is, compliance to all mandatory requirements and to the options that are not supported are shown in the completed DPRL. The resulting completed DPRL is called a DSPICS. Where this profile refines the features of the base standards, the requirements expressed in this DPRL shall be applied (as indicated in DPRL items with no "Profile Support" column) to constrain the allowable responses in the base standard DPICS proforma. When this profile makes additional requirements, the "Profile Support" column for such DPRLs shall be completed. In this column, each response shall be selected either from the indicated set of responses, or it shall comprise one or more parameter values as requested. If a conditional requirement is inapplicable, use the Not Applicable (NA) choice. If a

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the currentversion before use.

mandatory requirement is not satisfied, exception information must be supplied by entering a reference Xi, where i is a unique identifier, to an accompanying rationale for the noncompliance. When the profile requirement is expressed as a two-character combination (as defined in A.1.1 above), the response shall address each element of the requirement; e.g., for the requirement "mo," the possible compliant responses are "yy" or "yn."

A.2 Standards Referenced

This profile specifies the provision of TP0 service in an End System on top of a standardized TCP transport service. It uses the following standards:

RFC-1006	*(1987, May, ISO Transport Service on Top of The TCP, Version:3)*
ISO/IEC 8073	*(1992-12-15, Information technology Telecommunications and information exchange between systems- Open Systems Interconnection- Protocol for providing the connection-mode transport service)*
ISO/IEC 8072	*(1986, Information processing systems- Open systems Interconnection- Transport service definition)*
MIL-STD-2045-14502	*Version 1.0, 1994-02-10 (Information Technology - Defense Standardized Profile (DSP) - Internet Transport Profiles - Part 1: Transport Services)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:40Z
Check the source to verify that this is the current version before use.

A.3 DSPICS Requirements List

A.3.1 General Information

A.3.1.1 Implementation Identification

Ref	Question	Response
1	Date of Statement (DD/MM/YY)	
2	PICS serial number	
3	System Conformance statement cross-reference	

A.3.1.2 Identification of IUT

Ref	Question	Response
1	Implementation name	
2	Implementation version	
3	Machine name	
4	Machine version	
5	Operating System name	
6	Operating System version	
7	Special Configuration	
8	Other Information	

A.3.1.3 Global Statement of Conformance

Ref	Question	Response
1	Are all mandatory requirements met for:	
2	RFC 1006	
3	MIL-STD-2045-14502	
4	ISO/IEC 8072	
5	ISO/IEC 8073	

A.4 Basic Requirements

The following table lists the detailed requirements for an TP0/IP implementation.

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the current version before use.

Index	Protocol	Clause of Profile	Profile Status	Support
TCP	MIL-STD-2045-14502	5.2.1	m	Yes
TP0	ISO/IEC 8073	5.2.3	m	Yes
TP0	RFC 1006	5.2.2	m	Yes

A.4.1 PICS Proforma for TP0 (modified IAW RFC 1006)

A.4.1.1 Identification

Supplier	
Contact Point for Queries	
Implementation name(s) and Versions	
Other Information Necessary for full identification - eg, name(s) and version(s) for machines and/or operating systems; System Name(s)	

A.4.1.2 Protocol Summary

Protocol Version	
Addenda Implemented	
Amendments Implemented	
Date of Statement	

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:40Z
Check the source to verify that this is the current version before use.

A.4.1.3 Classes Implemented

Index	class	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
CO	CLASS 0	ISO/IEC 8073	M	5.2.3	m	Yes

A.4.1.3.1 Initiator /responder capability for protocol classes 0-4

Index		Base Standard		Profile		Support
		Reference	Status	Clause	Status	
IR1	initiating CR TPDU	ISO/IEC 8073	M	5.2.3	m	Yes
IR2	Responding to CR TPDU	ISO/IEC 8073	M	5.2.3	m	Yes

A.4.1.4 Supported Functions for class 0 (C0::)

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
TOF1	assignment to network connecton when operating over CONS	ISO/IEC 8073	M	5.2.3.2.1	m	Yes
TOF2	Port Addressing (106)	ISO/IEC 8073	M		m	Yes
TOF3	segmenting	ISO/IEC 8073	M		m	Yes
TOF4	reassembling	ISO/IEC 8073	M		m	Yes
TOF5	reassembling	ISO/IEC 8073	M		m	Yes
TOF6	connection refusal	ISO/IEC 8073	M		m	Yes
TOF7	normal release when operating over CONS	ISO/IEC 8073, RFC 1006 sec 5	M		m	Yes
TOF8	association of TPDUs with transport connection when operationg over CONS	ISO/IEC 8073	M		m	Yes

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the currentversion before use.

Item	Protocol Feature	Base Standard		Profile	Support
TOF9	Association of TPDUs with Transport connection when operationg over CONS	ISO/IEC 8073	M	m	Yes
TOF10	Treatment of protocol errors when operating over CONS	ISO/IEC 8073	M	m	Yes

A.4.1.4.1 Supported TPDUs

Index	TPDUs		Base Standard		Profile		Support
			Reference	Status	Clause	Status	
ST1	CR	supported on transmission	ISO/IEC 8073, 13.1	M	5.2.3.2.3.1	m	Yes
ST2	CR	supported on receipt	ISO/IEC 8073, 13.1	M	5.2.3.2.3.1	m	Yes
ST3	CC	supported on transmission	ISO/IEC 8073, 13.1	M	5.2.3.2.3.2	m	Yes
ST4	CC	supported on receipt	ISO/IEC 8073, 13.1	M	5.2.3.2.3.2	m	Yes
ST5	DR	supported on transmission	ISO/IEC 8073, 13.1	M	5.2.3.2.3.3	m	Yes
ST9	DT	supported on transmission	ISO/IEC 8073, 13.1	M	5.2.3.2.3.4	m	Yes
ST11	ED	supported on transmission	ISO/IEC 8073, 13.1, rfc 1006	C	5.2.3.2.3.5	m	Yes
ST19	ER	supported on receipt	ISO/IEC 8073, 13.1	M	5.2.3.2.3.6	m	Yes

A.4.1.4.2 Support of ER on transmission

Index	Class	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
ST1	Class 0	ISO/IEC 8073, 6.22.1	O	5.2.2.3.6	O	Yes/No

A.4.1.4.3 Supported parameters for class 0 TPDUs

Index	Supported parameters	Base Standard		Profile		Support
		Reference	Status	Clause	Status	

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the current version before use.

10CR6	Called TSAP-ID	ISO/IEC 8073 ,13.3.4a)	O	5.2.3.2.3.1	O	YES/NO
10CR7	Calling TSAP-ID	ISO/IEC 8073 ,13.3.4a)	O	5.2.3.2.3.1	O	YES/NO
10CR8	TPDU Size	RFC 1006 sec 5	M	5.2.3.2.3.1	M	YES/NO
10CR9	Preferred maximum TPDU size	ISO/IEC 8073 ,13.3.4c)	O	5.2.3.2.3.1	O	YES/NO

1. TPDU size shall be 65531 octets

The following parameters are optional if a CC TPDU is issued in class 0

Index	Supported parameters	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
10CC6	Called TSAP-ID	ISO/IEC 8073 ,13.4.4	O	5.2.2.2.3.2	O	
10CC7	Calling TSAP-ID	ISO/IEC 8073 ,13.4.4	O	5.2.2.2.3.2	O	
10CC8	TPDU Size[1]	RFC 1006 sec 5	M	5.2.2.2.3.2	M	
10CC9	Preferred maximum TPDU size	ISO/IEC 8073 ,13.4.4	O	5.2.2.2.3.2	O	

1. *TPDU size shall be 65531 octets*

The following parameters are optional if a CR TPDU is issued with preferred class 1

Index	Supported parameters	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
10DR4	Additional information	ISO/IEC 8073,13.5.4a)	O	5.2.2.2.3.31	O	Yes/No

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the currentversion before use.

ANNEX B (informative)

CONCLUDING MATERIAL

B.1 Deviations from the Base Standards/Referenced profiles

This MIL-STD documents the interface between TP0 and TCP for using ISO applications on top of a TCP transport stack. The profile is documented in the ISO/IEC TR 10000, "Framework and Taxonomy of International Standardized Profiles" and MIL-HDBK-829 format. This DSP does not deviate from the protocol as written in the RFC base standards.

B.2 Subject Term (Key word) Listing

Communication Protocol Standards
Data Communications
DDN
Functional Profiles
Interoperability
Internet
ISO Applications
ISO
Network
Relay
Standards
TCP
TP0
Transport

B.3 Preparing Activity

DISA-JIEO (Project DTMP-0004)

B.4 Reviewing Activities

Army:	SC
Air Force:	90
DLA:	DH

B.5 Custodians

DISA:	DC
Army:	SC
Air Force:	90

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:40Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1,2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-STD-2045-14503	2. DOCUMENT DATE (YYMMDD) 940701

3. DOCUMENT TITLE **Information Technology - DoD Standardized Profiles - Internet Transport Services Supporting OSI Applications**

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY **DEFENSE INFORMATION SYSTEMS AGENCY (DISA)**

a. NAME Dr. Frank D. Curcio	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)* Director JIEO Attn: JEBBF Ft. Monmouth, NJ 07703-5613	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN289-2340

Source: https://assist.dla.mil -- Downloaded: 2026l0-24T01:40Z
Check the source to verify that this is the currentversion before use.